

THE CULPRIT

Architecture & Hosting

Cost

System architecture, request flow, caching and abuse protection, followed by the full monthly hosting cost breakdown and the reasoning behind it.

Hosting: self-hosted VPS, Docker

CDN/Edge: Cloudflare

Traffic: up to 100 visitors/day

CONTENTS

1. Architecture

System architecture · application layers · request flow · cache layers & TTLs · admin invalidation · rate limiting & Turnstile

2. Hosting Cost

The two options · recommendation · full monthly bill · server sizing · supplier pricing · risks

Architecture

Diagrams of the live system — request flow, caching, and abuse protection.

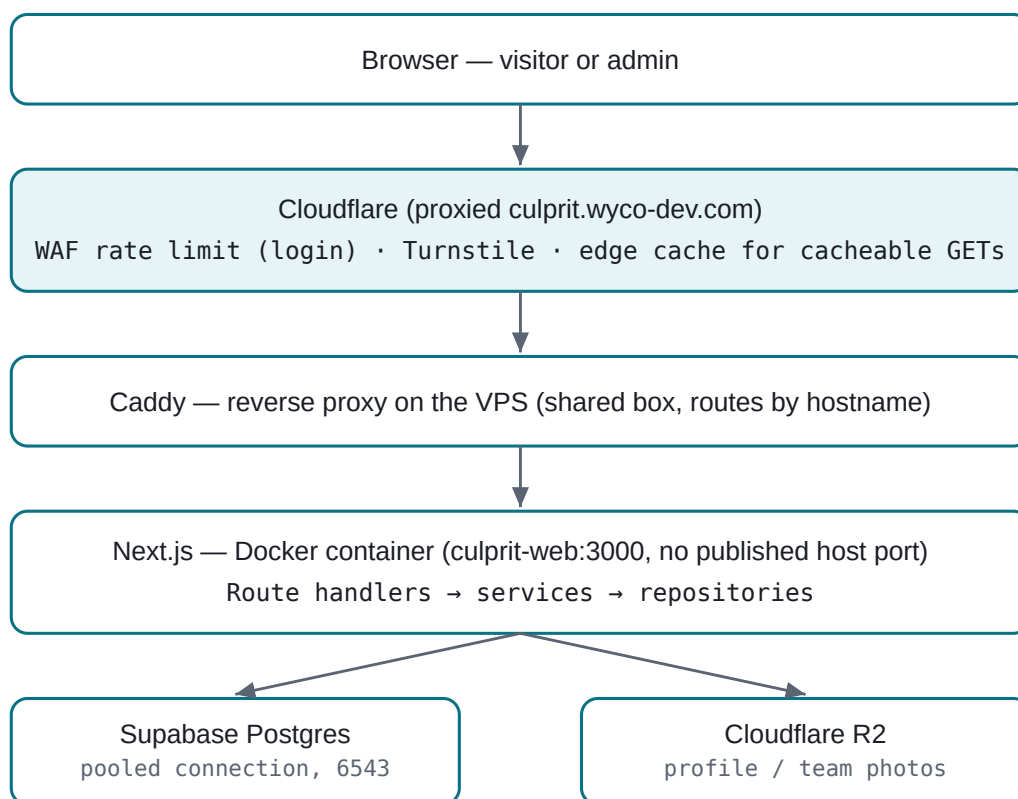
Hosting: self-hosted VPS, Docker

CDN/Edge: Cloudflare

Status: Live

01 System Architecture

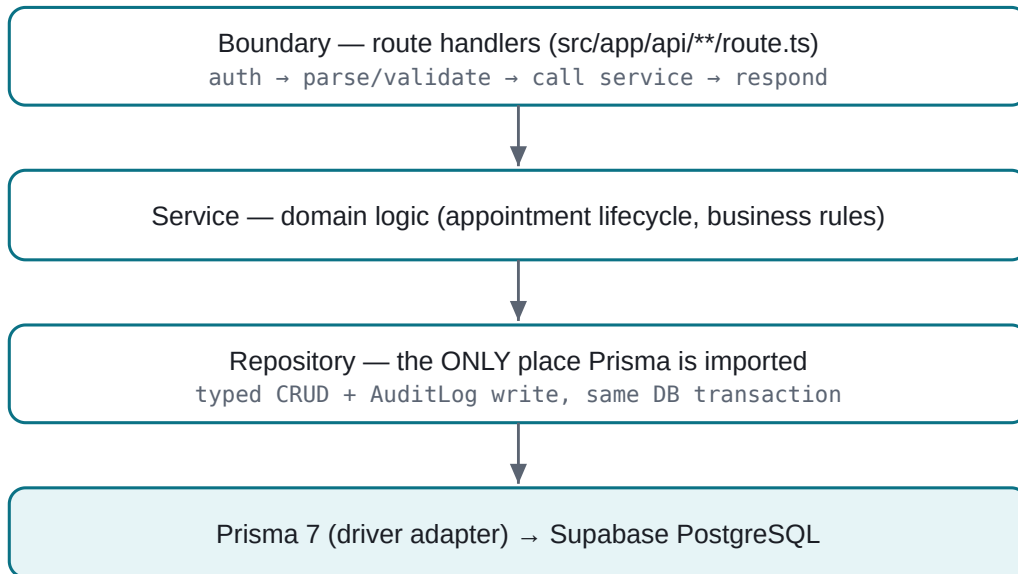
Every request — visitor or admin — enters through the same front door. There is no separate API host or second deployment: one Next.js container handles pages and API routes alike.



Calendly is embedded client-side only — the browser talks to Calendly directly; no server-side hop.

02 Application Layers

Inside the Next.js container, data flows one direction — each layer depends only on the one below it. Prisma is imported in exactly one place: the repository layer.

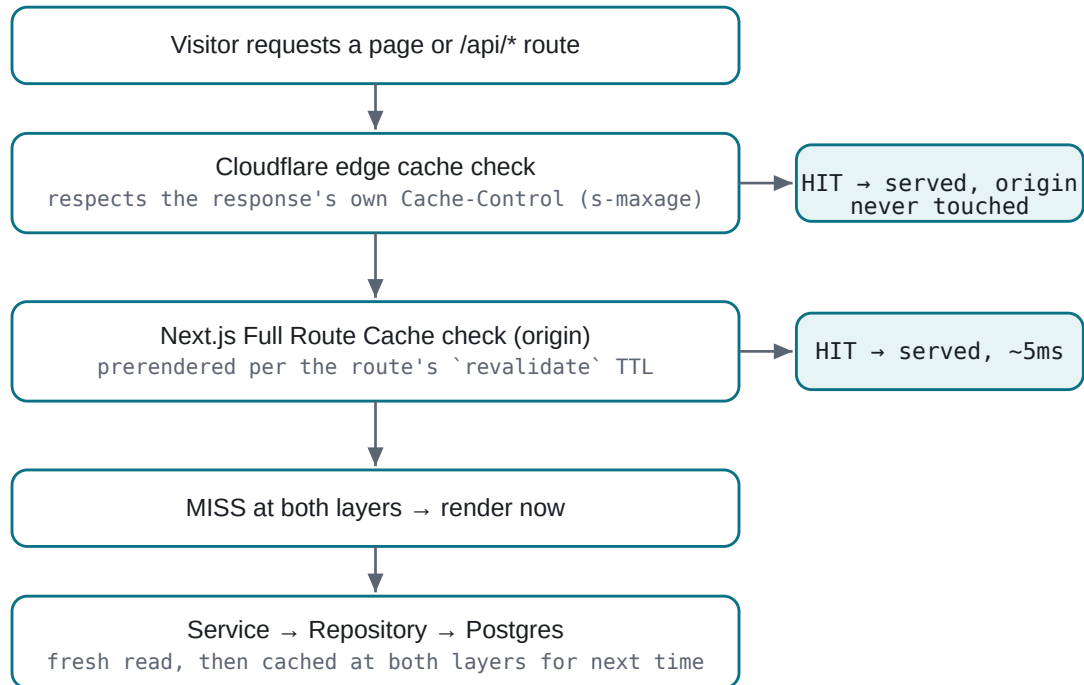


The audit-log write happens inside the repository, atomic with the mutation — not in the service layer.

Modules (`src/modules/*`): `auth` , `profile` , `research` , `publications` , `research-groups` , `appointments` , `settings` , `integrations` (Calendly embed, R2 storage, Turnstile, rate limiter, YouTube), `shared` (Result/error types, logger, API envelope, Prisma singleton).

03 How a Visitor Gets Data

A public page or API route can be answered from three different places, checked in order. Only a full cache miss at every layer actually touches the database.



CDN (Cloudflare edge) is the outermost layer; Next's Full Route Cache is the innermost, on the origin itself.

04 Cache Layers & TTLs

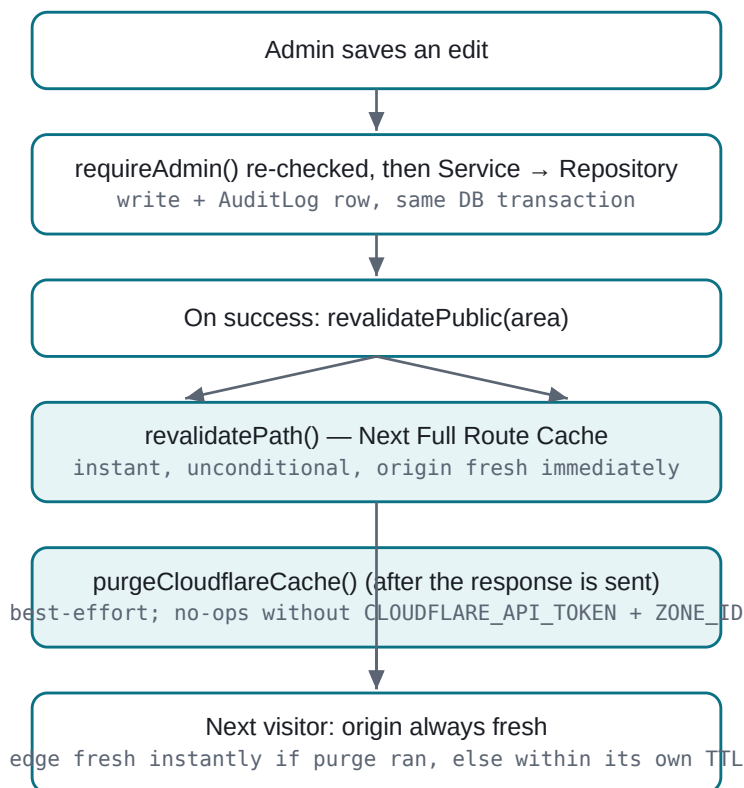
Real numbers from the running config — the browser and edge intentionally get different TTLs (`max-age` vs `s-maxage`), so a stale browser tab catches up sooner than the shared edge cache does.

CONTENT	BROWSER (MAX-AGE)	EDGE / ORIGIN (S-MAXAGE)
Static assets (JS, CSS, images)	1 year	1 year
Public pages (About, Research, etc. — Full Route Cache)	—	1 hour
Research / publications / profile / team-members API	5 min	1 hour
Upcoming-events API	1 min	5 min
Anything behind admin login (<code>/api/auth/*</code> , <code>/api/admin/*</code>)	no-store	no-store

Events gets a shorter TTL on purpose — an event can slip from "upcoming" to "past" while nobody's looking, so a shorter window limits how long a stale, already-past event can keep showing.

05 Admin: No Cache, Invalidate on Save

Admin pages and API routes are never cached — `requireAdmin()` re-checks the session on every single request, and the response is marked `private, no-store` explicitly (defense-in-depth, not just the absence of a header). A successful save then pushes freshness back out to visitors immediately.

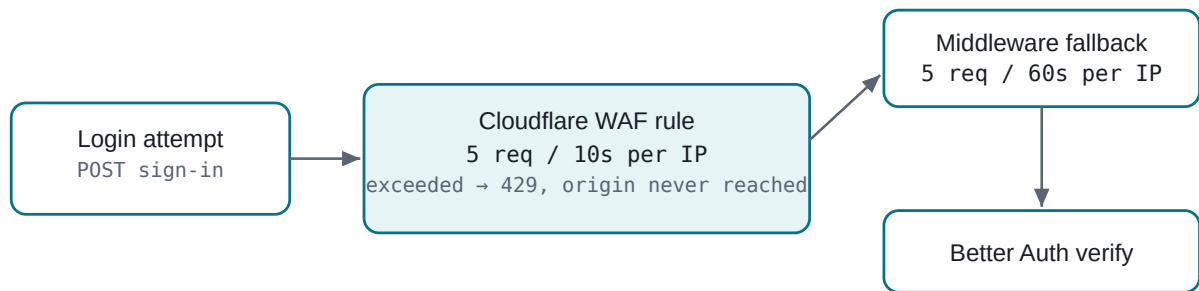


Origin freshness never depends on the Cloudflare token — it's a pure optimization on top of an already-correct origin.

06 Rate Limiting & Turnstile

Login brute force

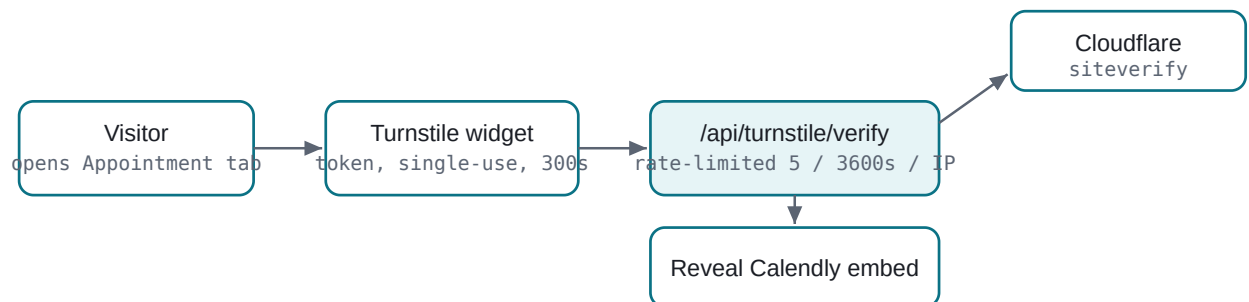
Two layers, because Cloudflare's Free plan allows exactly one WAF rate-limiting rule per zone — it's spent on the highest-severity target.



The middleware fallback (in-memory, per-instance) also covers mutating `/api/admin/*` at 30 req/60s – a stolen session cookie can't be hammered, though Better Auth's own session check is the real gate there.

Turnstile — gating the Calendly embed

Calendly's free plan has no bot protection and one bookable event type — Turnstile decides whether the booking widget appears at all.



The client-side widget proves nothing by itself – only a confirmed server-side siteverify call reveals the booking calendar. Failure keeps it hidden (fail-closed).

Hosting Cost

What it costs to run the site on a dedicated server, at the lowest realistic price.

Traffic: up to 100 visitors/day

Rate: \$1 = ₱33.10

Revision 4 · 12 Aug 2026

01 The Two Options

Both options run on a dedicated server, meaning nothing else shares it. The only real difference is where the database and files live.

OPTION	WHAT IT USES	COST
A Recommended	Dedicated server for the app, database on Supabase, files on Cloudflare R2. The same setup already running today, just moved to a server nothing else shares.	~\$7 ₱232/mo
B	Dedicated server running its own database and its own file storage, no Supabase, no R2	~\$13–14 ₱430–464/mo

02 Recommendation

Option A. The app itself is light on memory, so a small server is genuinely enough for it. Supabase and Cloudflare R2 are both still free at this site's size, and they already handle the harder parts (backups, serving files worldwide) better than a cheap server could do on its own.

Option B saves nothing and costs more. Running your own database means running your own backups too, and doing that properly (copying them somewhere safe, not just onto the same disk) takes real setup work. For a personal site with modest traffic, that's effort spent solving a problem Option A already has solved for free.

The one reason to still pick Option B: not wanting the database and files to depend on a company's free-tier terms staying the same. Reasonable to want, but it costs more, in both money and setup effort, to get it.

03 Full Monthly Bill

Option A: dedicated server + Supabase + Cloudflare R2

ITEM	PLAN	PER MONTH	PER YEAR
Server	DigitalOcean Basic, 1 vCPU / 1 GB RAM / 25 GB SSD	\$6.00 ฿199	\$72 ฿2,383
Domain	One domain, address and email	\$1.00 ฿33	\$12 ฿397
Database	Supabase Free	\$0	\$0
Photos and files	Cloudflare R2 Free	\$0	\$0
CDN, HTTPS, attack protection	Cloudflare Free	\$0	\$0
Bot check	Cloudflare Turnstile	\$0	\$0
Booking calendar	Calendly Free, embedded	\$0	\$0
Total		\$7.00 ฿232	\$84 ฿2,780
Optional: no free-tier pause, daily backups	Supabase Pro	\$25.00 ฿828	\$300 ฿9,930

Supabase's free plan takes no automatic backups and pauses a project after a week with no traffic, until someone resumes it by hand. That's true regardless of which server hosts the app. Fixing it costs \$25/mo for Supabase Pro, on top of the \$7 above.

Option B: dedicated server, own database and storage

ITEM	PLAN	PER MONTH	PER YEAR
Server	DigitalOcean Basic, 1 vCPU / 2 GB RAM / 50 GB SSD	\$12.00 ฿397	\$144 ฿4,767
Domain	One domain, address and email	\$1.00 ฿33	\$12 ฿397
Database	Self-hosted Postgres, same server	Included	Included
Photos and files	Stored on the server's own disk	Included	Included
CDN, HTTPS, attack protection	Cloudflare Free	\$0	\$0
Bot check	Cloudflare Turnstile	\$0	\$0
Booking calendar	Calendly Free, embedded	\$0	\$0
Total		\$13.00 ฿430	\$156 ฿5,164

2 GB RAM is the realistic floor here, not 1 GB (see the sizing check below). Backups aren't included in this price; they'd need to be built separately (a scheduled export, copied somewhere other than the same disk), and there's no free service standing in for that the way Supabase does in Option A.

04 Server Sizes

PLAN	SPECIFICATION	TRANSFER	PER MONTH	FITS
Basic	1 vCPU · 512 MB RAM · 10 GB SSD	500 GB	\$4.00 ฿132	Too small, see below
Basic	1 vCPU · 1 GB RAM · 25 GB SSD	1,000 GB	\$6.00 ฿199	Option A pick
Basic	1 vCPU · 2 GB RAM · 50 GB SSD	2,000 GB	\$12.00 ฿397	Option B pick
Basic	2 vCPU · 2 GB RAM · 60 GB SSD	3,000 GB	\$18.00 ฿596	More than either option needs today

Kimi K3 is now available →

BlogDocsCareersGet SupportContact Sales

DigitalOcean

ProductsSolutionsDevelopersPartnersPricing

Log inSign up

← Back to overview

Basic

- CPU-Optimized
- General Purpose
- Memory-Optimized
- Storage-Optimized
- Additional product pricing
- FAQ

already reducing spending on short-lived workloads like running batch jobs and automated testing.

View our [DigitalOcean Droplet product documentation](#) to learn which Droplet plan is right for you or [talk to an expert](#).

Choose the right Droplet type

Basic Droplets

Basic Droplets have the most efficient CPU usage at a lower cost for workloads that underuse dedicated threads. They're ideal for bursty applications that can handle variable levels of CPU.

CPU Options

RegularPremium IntelPremium AMD

Memory	vCPU	Transfer	SSD	\$/hr	\$/mo
512 MiB	1 vCPU	500 GiB	10 GiB	\$0.00595	\$4.00
1 GiB	1 vCPU	1,000 GiB	25 GiB	\$0.00893	\$6.00
2 GiB	1 vCPU	2,000 GiB	50 GiB	\$0.01786	\$12.00
2 GiB	2 vCPUs	3,000 GiB	60 GiB	\$0.02679	\$18.00
4 GiB	2 vCPUs	4,000 GiB	80 GiB	\$0.03571	\$24.00

Screenshot: DigitalOcean, Basic plan, Regular (SSD) CPU option.

Other machine options at the same sizes

The same RAM/disk sizes are also available on two pricier processor types, **Premium AMD** and **Premium Intel**, which trade a higher price for faster NVMe storage:

CPU OPTION	~1 GB PLAN	~2 GB PLAN	DISK	WORTH IT HERE?
Regular	\$6.00 ¥199	\$12.00 ¥397	SSD	Yes , used above
Premium AMD	~\$7 ~¥232	~\$14 ~¥463	NVMe SSD	Faster disk, no benefit for a site this size
Premium Intel	\$8.00 ¥265	\$16.00 ¥530	NVMe SSD	Faster disk, no benefit for a site this size

This site serves text and small images, not large files or heavy database queries, so the extra disk speed is never the bottleneck. Regular keeps the bill as low as possible without giving anything up.

05 Can These Sizes Actually Run It? Checked, not assumed: against the app's own documented memory settings and general server-sizing guidance.

Option A's server (1 GB)

The app's own production settings already cap it at 512 MB of memory (`docker-compose.production.yml` : `NODE_OPTIONS=--max-old-space-size=256` , container limit `512m`), but that cap is for the app *container*, not the whole machine. The operating system, the Docker engine itself, and the Caddy reverse proxy in front of it also need memory, and DigitalOcean's own guidance recommends at least 1 GB of RAM to run Docker reliably. A 512 MB server (the \$4 plan) would leave next to nothing for that overhead, which is why it's marked "too small" above rather than recommended, even though it's cheaper. The 1 GB plan gives the app its documented 512 MB and leaves roughly the other half for the system around it, which matches DigitalOcean's own floor.

Option B's server (2 GB)

Postgres itself can run in very little memory for a small, low-traffic, single-user database (some guides cite 256 MB as workable with a tuned-down configuration). That's Postgres alone, though. Add the app's own 512 MB and the same OS/Docker overhead from above, and a 1 GB server leaves barely any margin: fine most of the time, but a bad moment for the app and the database to both be doing something at once could tip it into swapping or an out-of-memory restart. A 2 GB server gives each piece real headroom instead of a size that technically starts but stays fragile, which is why it's the pick above rather than the cheaper 1 GB plan.

Neither pick is guesswork: both are sized against the app's own configuration file and DigitalOcean's own minimum-RAM guidance for Docker, not just "smallest available."

06 Supplier Pricing

Supabase (Option A's database)

PLAN	PER MONTH	INCLUDED	CATCH
Free In use today	\$0	500 MB database, 5 GB downloads/month	No backups, pauses after 7 idle days
Pro	\$25 ฿828	8 GB database, daily backups, no pause rule	Optional upgrade

Product ▾Developers ▾Solutions ▾PricingDocsBlog

107.6KSign inStart your project

Predictable pricing, designed to scale

Start building for free, collaborate with your team, then scale to millions of users

FREE

Perfect for passion projects & simple websites.

Start for Free

\$0 / month

Get started with:

✓ Unlimited API requests

✓ 50,000 monthly active users

✓ 500 MB database size
Shared CPU • 500 MB RAM

✓ 5 GB egress

✓ 5 GB cached egress

✓ 1 GB file storage

✓ Community support

PRO Most Popular

For production applications with the power to scale.

Get Started

From \$25 / month

First project included. Additional projects from \$10/mo. See how pricing scales

Everything in the Free Plan, plus:

✓ 100,000 monthly active users then \$0.00325 per MAU

✓ 8 GB disk size per project then \$0.125 per GB

✓ 250 GB egress then \$0.09 per GB

✓ 250 GB cached egress then \$0.03 per GB

✓ 100 GB file storage

TEAM

Add features such as SSO, control over backups, and industry certifications.

Get Started

From \$599 / month

First project included. Additional projects from \$10/mo. See how pricing scales

Everything in the Pro Plan, plus:

✓ SOC2 & ISO 27001

✓ Project-scoped and read-only access

✓ HIPAA available as paid add-on

✓ SSO for Supabase Dashboard

✓ Priority email support & SLAs

✓ Daily backups stored for 14 days

ENTERPRISE

For large-scale applications running Internet scale workloads.

Contact Us

Custom

✓ Designated Support manager

✓ Uptime SLAs

✓ BYO Cloud supported

✓ 24×7×365 premium enterprise support

✓ Private Slack channel

✓ Custom Security Questionnaires

Screenshot: `supabase.com/pricing`. Free tier is \$0; this site's data is well under the 500 MB limit.

Cloudflare R2 (Option A's file storage)

PLAN	PER MONTH	INCLUDED
Free In use today	\$0	10 GB storage, unlimited free downloads

CLOUDFLARE

DOCS

DirectoryAPISDKsChangelog

Search

CtrlK

Log in

R2

Filter sidebar

Overview

Get started

How R2 works

Data migration

Buckets

Objects

API

R2 Data CatalogBeta

R2 SQLBeta

Examples

Tutorials

Videos

Demos and architectures

Platform

Reference

Pricing

Agent resources

To learn about potential cost savings from using R2, refer to the [R2 pricing calculator](#).

R2 pricing

	Standard storage	Infrequent Access storage
Storage	\$0.015 / GB-month	\$0.01 / GB-month
Class A Operations	\$4.50 / million requests	\$9.00 / million requests
Class B Operations	\$0.36 / million requests	\$0.90 / million requests
Data Retrieval (processing)	None	\$0.01 / GB
Egress (data transfer to Internet)	Free ¹	Free ¹

⚠ Billable unit rounding

Cloudflare rounds up your usage to the next billing unit.

For example:

- If you have performed one million and one operations, you will be billed for two million operations.
- If you have used 1.1 GB-month, you will be billed for 2 GB-month.
- If you have retrieved data (for infrequent access storage) for 1.1 GB, you will be billed for 2 GB.

Free tier

You can use the following amount of storage and operations each month for free.

	Free
Storage	10 GB-month / month
Class A Operations	1 million requests / month
Class B Operations	10 million requests / month

On this page

Overview

R2 pricing

Free tier

Storage usage

Class A operations

Class B operations

Free operations

Data retrieval

Minimum storage duration

R2 Data Catalog pricing

Data migration pricing

Super Slurper

Sippy

Pricing calculator

R2 billing examples

Standard storage example

Infrequent access example

Asset hosting

Cloudflare billing policy

Frequently asked questions

Will I be charged for unauthorized requests to my R2 bucket?

WAS THIS HELPFUL?

YesNo

Edit page

Screenshot: `developers.cloudflare.com/r2/pricing`. Sending files to visitors is free on every plan.

Everything else (both options)

Cloudflare (web address, worldwide caching, HTTPS, attack protection, Turnstile bot check) and Calendly (embedded booking widget) are free on both options and unaffected by the hosting choice. The domain is about \$12/year (฿397) either way.

07 Risks

RISK	OPTION A	OPTION B
Database has no backup	True on the free plan , fixed for \$25/mo	True until built by hand
Database pauses if quiet a week	True on the free plan	Doesn't happen
Server runs low on memory under load	Unlikely , sized with headroom	Possible at the 1 GB tier, why 2 GB is the pick
Losing the server loses everything	No , database and files live elsewhere	Yes , unless backups are copied off the server too
Setup and ongoing effort	Low , mostly already done	Higher , someone maintains the database

The honest read: Option A is cheaper and safer by default. Option B only earns its extra cost and effort if not depending on Supabase or Cloudflare's free tiers matters more than money or convenience.